LESSON PLAN

ADVANCED AIRCRAFT TECHNOLOGY (20)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	05
Total No. of Lesson Plans	20	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction to ATE and its uses
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction to ATE and its uses.
5. Specific Objectives	20.2.1.1 Trainee will be able to analyze the uses of ATE.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of ATE definitions and applications; Safety: No live equipment handling; Supervision: Instructor monitors understanding
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What does the acronym ATE stand for?
(b) Can you list two traditional methods of aircraft maintenance?
(c) Why is technology integration necessary in modern aviation?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of ATE • Core components of ATE • Data acquisition systems • Signal processing basics • Integration with aircraft systems	L & Ds	PPT, IB	15 min
(b)	• Predictive maintenance capabilities • Fault isolation techniques • Condition monitoring applications • Cost reduction benefits • Operational efficiency gains	L & Ds	PPT, PDF	20 min
(c)	• Future trends in ATE • AI and machine learning roles • Cybersecurity considerations • Regulatory compliance standards • Training implications for aircrew	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) How does ATE differ from traditional reactive maintenance?
(b) Explain one specific use of ATE in engine health monitoring.
(c) What is the primary benefit of using ATE for fault isolation?

12. Home Assignment and Follow Up	Prepare a brief note on how ATE principles introduced in this lesson will be applied in the study of electronic instruments in LP 06.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Introduction to electronic instruments and advantages.	Refer to the master syllabus reference list	LP 06

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min